

LuxuryStay HMS – Roman English Viva Practice Sheet

1. PROJECT INTRODUCTION

Q: Apne project ke bare mein batao.

Sir, hamara project 'LuxuryStay Hospitality – Hotel Management System' hai. Ye aik complete MERN stack web application hai. Iska maqsad hotel ki bookings, rooms, aur staff ko easily manage karna hai. Ismein guest rooms book kar sakte hain, aur admin dashboard se poore hotel ka revenue, staff ke tasks aur check-ins ko handle kar sakta hai. Ye manual register system ko khatam karke sab kuch digital aur fast banata hai.

2. TECHNOLOGIES USED IN OUR PROJECT

Q1: React kya hai?

Simple Roman English Answer:

React aik JavaScript library hai jo hum frontend aur user interface banane ke liye use karte hain.

Hamare Actual Project Code Ki Example:

```
import React, { useState } from 'react';
function App() {
  return <div>Welcome to LuxuryStay</div>;
}
```

Code Ki Roman English Explanation:

Ye ek basic React component hai jo browser par 'Welcome to LuxuryStay' show karega.

Viva Mein Bolne Wala Short Answer:

Sir, React humne apni website ka frontend banane ke liye use ki hai taake website bina reload hue fast chale.

Related File/Folder:

views/luxury-stay-frontend/src/App.jsx

Q2: Node.js kya hai?

Simple Roman English Answer:

Node.js aik runtime environment hai jo JavaScript ko browser ke bahar server par run karne ki ijazat deta hai.

Hamare Actual Project Code Ki Example:

```
const express = require('express');
const app = express();
app.listen(5000, () => console.log('Server running on port 5000'));
```

Code Ki Roman English Explanation:

Yahan Node.js ki madad se humne backend ka server start kiya hai jo port 5000 par run ho raha hai.

Viva Mein Bolne Wala Short Answer:

Sir, Node.js humara backend server run karne ke liye use hota hai.

Related File/Folder:

server.js

Q3: MongoDB kya hai?

Simple Roman English Answer:

MongoDB aik NoSQL database hai jo data ko tables ke bajaye JSON (documents) format mein save karta hai.

Hamare Actual Project Code Ki Example:

```
mongoose.connect(process.env.MONGO_URI, {  
  useNewUrlParser: true  
});
```

Code Ki Roman English Explanation:

Ye code hamare Node backend ko MongoDB database se connect kar raha hai.

Viva Mein Bolne Wala Short Answer:

Sir, MongoDB humara database hai jahan hotel ke users aur bookings ka data save hota hai.

Related File/Folder:

config/db.js ya server.js

Q4: Express.js kya hai?

Simple Roman English Answer:

Express.js Node.js ka aik framework hai jo APIs (routes) banane ko bohot asaan kar deta hai.

Hamare Actual Project Code Ki Example:

```
const router = express.Router();  
router.get('/rooms', getRooms);
```

Code Ki Roman English Explanation:

Yahan Express router use ho raha hai jo '/rooms' ke URL par getRooms ka function call karega.

Viva Mein Bolne Wala Short Answer:

Sir, Express.js backend mein APIs banane aur URLs ko manage karne ke liye use hota hai.

Related File/Folder:

routes/roomRoutes.js

Q5: Tailwind CSS kya hai?

Simple Roman English Answer:

Tailwind aik utility-first CSS framework hai jisse hum classes directly HTML/JCX mein likh kar styling karte hain.

Hamare Actual Project Code Ki Example:

```
<button className='bg-blue-500 text-white p-2 rounded'>
  Book Now
</button>
```

Code Ki Roman English Explanation:

Is code mein 'bg-blue-500' se button ka color blue aur 'text-white' se text white ho raha hai bina alag se CSS file banaye.

Viva Mein Bolne Wala Short Answer:

Sir, Tailwind CSS humne apni website ka design aur styling jaldi aur khoobsurat banane ke liye use ki hai.

Related File/Folder:

Frontend Components

3. FRONTEND QUESTIONS

Q6: useState kya hai?

Simple Roman English Answer:

useState React ka aik hook hai jo component ke andar data (state) ko save aur update karne ke liye use hota hai.

Hamare Actual Project Code Ki Example:

```
const [isSaving, setIsSaving] = useState(false);
// To update:
setIsSaving(true);
```

Code Ki Roman English Explanation:

Yahan 'isSaving' aik variable hai aur 'setIsSaving' usko update karne ka function. Shuru mein iski value false hai.

Viva Mein Bolne Wala Short Answer:

Sir, useState screen par data store aur change karne ke liye use hota hai.

Related File/Folder:

views/luxury-stay-frontend/src/pages/Rooms.jsx

Q7: useEffect kya hai?

Simple Roman English Answer:

useEffect React ka hook hai jo tab run hota hai jab page load hota hai (ya data change hota hai), jaise API se data mangwana.

Hamare Actual Project Code Ki Example:

```
useEffect(() => {
  fetchRooms();
}, []);
```

Code Ki Roman English Explanation:

Jab bhi ye page khulega, ye code automatically backend se rooms ka data (fetchRooms) mangwayega.

Viva Mein Bolne Wala Short Answer:

Sir, useEffect page load hotay hi backend se data fetch karne ke liye use hota hai.

Related File/Folder:

Frontend Pages

Q8: Axios kya hai?

Simple Roman English Answer:

Axios aik library hai jo frontend se backend tak requests (GET, POST) bhejney ke liye use hoti hai.

Hamare Actual Project Code Ki Example:

```
const response = await axios.post('/api/auth/login', { email, password });
```

Code Ki Roman English Explanation:

Yahan frontend se user ka email aur password backend ki login API par bheja ja raha hai.

Viva Mein Bolne Wala Short Answer:

Sir, Axios React se Node backend ko data bhejne aur mangwane ke liye use hota hai.

Related File/Folder:

Frontend api.js

4. BACKEND QUESTIONS

Q9: Mongoose kya hai?

Simple Roman English Answer:

Mongoose MongoDB ke liye aik library hai jo data ka structure (schema) define karne mein madad karti hai.

Hamare Actual Project Code Ki Example:

```
const RoomSchema = new mongoose.Schema({
  roomNumber: { type: String, required: true },
  price: Number
});
```

Code Ki Roman English Explanation:

Yahan hum bata rahe hain ke Room ke data mein roomNumber string hoga aur price number hoga.

Viva Mein Bolne Wala Short Answer:

Sir, Mongoose database mein data ka structure aur rules banane ke liye use hota hai.

Related File/Folder:

Model/Room.js

Q10: Middleware kya hota hai?

Simple Roman English Answer:

Middleware aik function hai jo request anay aur controller tak pohncne ke darmiyan run hota hai (jaise security check).

Hamare Actual Project Code Ki Example:

```
const protect = (req, res, next) => {  
  if(req.headers.authorization) {  
    next(); // Pass to controller  
  }  
};
```

Code Ki Roman English Explanation:

Ye code check kar raha hai ke user ke paas token hai ya nahi, agar hai tou 'next()' call hoga.

Viva Mein Bolne Wala Short Answer:

Sir, Middleware API ko protect karne aur request ko verify karne ke liye use hota hai.

Related File/Folder:

middleware/auth.js

5. AUTHENTICATION & SECURITY

Q11: JWT (JSON Web Token) kya hai?

Simple Roman English Answer:

JWT aik secure token hai jo user ko login hone par milta hai taake wo agli dafa bina password ke verify ho sakay.

Hamare Actual Project Code Ki Example:

```
const token = jwt.sign({ id: user._id }, process.env.JWT_SECRET, {  
  expiresIn: '30d'  
});
```

Code Ki Roman English Explanation:

Yahan user ki ID ko secret key ke sath lock kar ke 30 din ka token banaya ja raha hai.

Viva Mein Bolne Wala Short Answer:

Sir, JWT login ke baad user ko verify rakhne ka aik secure tareeqa hai.

Related File/Folder:

controllers/authController.js

Q12: bcrypt kya hai?

Simple Roman English Answer:

bcrypt aik library hai jo user ke password ko encrypt (hash) karke database mein save karti hai taake koi password parh na sakay.

Hamare Actual Project Code Ki Example:

```
const isMatch = await bcrypt.compare(enteredPassword, user.password);
```

Code Ki Roman English Explanation:

Ye code user ke dale hue password ko database mein mojud encrypted password se match kar raha hai.

Viva Mein Bolne Wala Short Answer:

Sir, bcrypt passwords ko secure aur hash karne ke liye use hota hai.

Related File/Folder:

Model/User.js / authController.js

6. PROJECT MODULES

Q13: Room Management module kya karta hai?

Simple Roman English Answer:

Is module mein Admin naye rooms add kar sakta hai, unki price set kar sakta hai aur unhe delete ya update kar sakta hai.

Hamare Actual Project Code Ki Example:

```
const createRoom = async (req, res) => {  
  const room = await Room.create(req.body);  
  res.status(201).json(room);  
};
```

Code Ki Roman English Explanation:

Ye backend function database mein naya room create kar raha hai aur 201 Success status bhej raha hai.

Viva Mein Bolne Wala Short Answer:

Sir, is module ke zariye admin hotel ke total rooms aur unki details ko control karta hai.

Related File/Folder:

controllers/roomController.js

7. SIMPLE PROJECT WORKFLOW

Technical Workflow:

Frontend (React) --> API Request (Axios) --> Backend (Express) --> Controller/Logic --> Database (MongoDB) --> Response Back to Frontend

Hotel Workflow:

Guest Login karta hai --> Room search karta hai --> Booking confirm karta hai --> Admin Dashboard mein booking dekhta hai --> Guest check-in and check-out karta hai.

8. IMPORTANT FILES & FOLDERS

File / Folder	Kya kaam karta hai	Actual Example
server.js	Backend ka main entry point, server start karta hai.	app.listen(5000)
.env	Passwords aur secret keys ko secure rakhta hai.	MONGO_URI=...
Model/	Database schemas define karta hai.	User.js, Room.js

routes/	APIs ke URLs banata hai.	router.post('/login')
controllers/	Asal logic likhi hoti hai route hit hone par.	authController.js
views/luxury-stay-frontend	React ka poora frontend isme hai.	App.jsx

9. QUICK REVISION SHEET

Project Name:

LuxuryStay Hospitality – Hotel Management System

Technologies:

React (UI), Node.js (Server), Express (API), MongoDB (Database), Tailwind (CSS), JWT (Auth).

Main Modules:

Admin Dashboard, Booking/Reservations, Room Management, Staff Portal.

Authentication:

JWT (token) aur bcrypt (password hashing).

API Flow:

Frontend -> Routes -> Middleware -> Controller -> Model -> DB.

Important Tip for Viva:

Agar teacher pooche MERN kyun use kiya? Tou bolna: 'Sir kyunki ye complete JavaScript stack hai, frontend aur backend dono mein JS use hoti hai jisse development fast aur asaan ho jati hai.'